Machine-checkable ecological evidence: replay-verified computation over biological connectomes in a consent-gated sensing commons

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Abstract

Ecological monitoring and biodiversity-credit markets depend on evidence that a sensor reading was produced where and how it is claimed. Existing approaches – trusted firmware, third-party auditors, and zero-knowledge or trusted-execution proof systems – either require field hardware trust or carry heavy proof-generation cost. We present an alternative: *replay-verified computation*, in which a deterministic simulator executed over a preserved biological wiring diagram produces a witness signature that any independent implementation can reproduce bit-for-bit, and *WiredReplay-1*, a conformance corpus of fifteen packed connectomes (cellular electron-microscopy and mesoscale parcellation) with canonical inputs and expected digests for validating third-party simulators. The witnesses sit inside a consent-gated sensing commons that separates evidence-gated lanes (sensing, fleet aggregation) from consent-gated lanes (emission to and storage on living systems) and refuses the rest as first-class protocol values. We demonstrate: (i) fifteen of fifteen connectomes relay a shared feature-vector input with fifteen topology-unique signatures, replayed identically by independent Python and TypeScript implementations (930/930 parity checks); (ii) at the largest cellular wiring, the packed simulator uses $\sim 10^{3}\times$ fewer estimated joules than an equivalent dense fp32 multiply on the same host, while a degree-matched shuffled control shows the saving comes from sparse packed representation, not biological topology – a negative result we report deliberately; (iii) the full protocol is exercised by 121 end-to-end checks and a live public deployment (71/71 checks) exposing content-addressed, receipt-gated verdicts. A small-wiring benchmark where our approach *loses* (Celegans, 302 neurons) is reported alongside the win. The corpus, specification, harnesses, and claim-evidence table are open

(AGPL/CC-BY), pre-registered in the accompanying methods repository, and intended as a citable standard for deterministic biologically-derived computation.

Keywords: connectome, spiking neural simulation, determinism, replay verification, provenance, ecological monitoring, MRV, consent, open science

1 Introduction

Ecological sensing is scaling faster than the evidence discipline that would make its outputs trustworthy. Autonomous acoustic recorders, eDNA samplers, and community-sensor networks now generate continuous streams of claimed observations about living systems Hill et al (2018); Sueur and Farina (2015); Kahl et al (2021); biodiversity-credit and conservation-finance markets are beginning to price claims derived from those streams Ford et al (2024). The evidentiary question — *was this reading produced as claimed, by what, and under whose authority?* — is typically answered by trusting firmware and auditors, or deferred to provenance standards that describe data lineage without constraining it W3C Provenance Working Group (2013); Wieczorek et al (2012).

We describe a sensing commons, *Temple Earth*, in which the evidence problem is treated as a verifiable-computation problem with an unusual verifier: preserved biological wiring diagrams (connectomes) replayed by deterministic simulation. The system is built around three separable contributions.

First, a **protocol** in which every capability is routed through one of two explicit gates. *Evidence-gated lanes* (sensing, fleet-wide aggregation) admit only inputs that carry a verifiable provenance claim. *Consent-gated lanes* (emission toward a living system, storage on a guardian-controlled node) admit only operations whose consent tuple — guardian, operator, action class, policy hash — verifies at admission time. Lanes that satisfy neither (including any computation claimed to run *inside* a living organism, and any relay that would write back into a sensed environment) are *refused as a first-class protocol value*: a signed refusal is itself a receipt, auditable and quotable. Emission, where permitted, is further constrained to minimal energy and duty-cycled bursts under joint guardian and operator consent — "the emission stack is consent-gated, minimal, and duty-cycled; the default posture is listen-only."

Second, **witnesses**: a deterministic leaky integrate-and-fire (LIF) simulator that runs the same packed graph step-for-step on a commodity host or in a cloud worker, producing a BLAKE2s digest of the full spike raster — the *witness signature* of that wiring diagram under that input. Because the simulation is deterministic by construction (fixed-point arithmetic, no floating-point scheduling, no RNG state), any conformant implementation reproduces the signature exactly; agreement between independent implementations is a cryptographic-strength coincidence, not a configuration detail. A quorum of witnesses — distinct connectomes agreeing on a verdict under quorum rules — folds into a single content-addressed evidence record (a *receipt*, addressed by self-authenticating identifier, SAID).

Third, **WiredReplay-1**: the corpus that makes the witness layer a shared standard rather than a proprietary implementation detail — fifteen packed wiring diagrams, canonical test vectors with expected digests, and the normative format and encoder specification.

1.1 Claims and evidence discipline

Every quantitative statement in this paper is labelled by the evidence tier used throughout the project: *proven by science* (published external literature), *built — heading outdoors* (implemented and tested, not field-deployed), *live now* (running in the deployed system), *measured*, *estimated*, *cited*, *projected*, or *refused*. The full claim table is Appendix C and the authoritative version is `docs/SCIENCE.md` in the repository. Where our own method performs poorly — and one such case is deliberately included — we report it.

1.2 Contributions

- A protocol architecture separating evidence-gated from consent-gated lanes, with signed refusal as a receipt (Section 3, Appendix A).
- Deterministic replay-verified computation over fifteen preserved biological connectomes, with independent implementations agreeing on every digest (Sections 4, 5).
- WiredReplay-1, an open conformance corpus: packed graphs, checksums, 45 canonical vectors, expected digests, and the encoder specification (Section 4.1).
- An honest efficiency analysis with a degree-matched shuffled control showing the measured gain is from sparse packed representation, not biological topology (Section 5.4).
- A threat model and claim-evidence table that bind every statement to its tier (Appendices B, C).

2 Related work

Connectomes at both resolutions. Cellular-scale wiring is now available for *C. elegans* (both sexes) White et al (1986); Cook et al (2019), tadpole *Ciona* Ryan et al (2016), *Drosophila* larva Ohyama et al (2015); Gerhard et al (2017) and adult (hemibrain Scheffer et al (2020), full FAFB/FlyWire Zheng et al (2018); Dorkenwald et al (2024)), *Platynereis* Verasztó et al (2020), and mouse retina Helmstaedter et al (2013); medulla-column data derive from Takemura et al. Takemura et al (2015). Mesoscale matrices exist for mouse Oh et al (2014), rat Bota et al (2015), macaque Markov et al (2014); Modha and Singh (2010), human Hagmann et al (2008), and cat Scannell and Young (1993). Prior work uses these graphs for circuit neuroscience; we use them as deterministic witnesses — a *representation* claim about reproducible computation, not a claim about cognition or ecology.

Simulation and reservoir computing. The LIF step is a standard discrete membrane equation Hodgkin and Huxley (1952); Brette et al (2007); Izhikevich (2003). Using recurrent wiring as an untrained readout substrate is reservoir computing Jaeger

Table 1 Verification approaches for remote environmental claims.

	proof cost	trust root	field cost
Trusted firmware/auditor	none	vendor + auditor	low
ZK / SNARK Groth (2016)	high (proving)	math	high
TEE Costan and Devadas (2016)	low	chip vendor	moderate
This work	**re-execution**	**determinism + open corpus**	**low**

(2001); Maass et al (2002); Lukoševičius and Jaeger (2009); OpenWorm demonstrates whole-animal simulation as an open-science artifact Szigeti et al (2014). Our contribution is not a simulator but a *verification discipline* around one: determinism requirements, a conformance corpus, and digest-level replay.

Neuromorphic baselines. TrueNorth Merolla et al (2014), Loihi Davies et al (2018), and SpiNNaker Furber et al (2014) run spiking networks at low power on dedicated silicon. We deliberately compare against same-host CPU baselines instead: the relevant question for an edge sensing commons is whether *this* representation is cheap on the hardware already deployed.

Verifiable computation. Cryptographic verification spans interactive proofs, succinct arguments Groth (2016), and trusted execution Costan and Devadas (2016); Walfish and Blumberg's survey framed the trade-off: verify without re-executing, at a price in proof cost or hardware trust Walfish and Blumberg (2015). Certificate Transparency showed the value of append-only, publicly replayable logs for a different domain Laurie et al (2014), and supply-chain frameworks (in-toto Torres-Arias et al (2019)) popularized receipted-build attestations. Our point of comparison: where proof systems sell *verification without re-execution*, we sell the opposite — *re-execution as the receipt* — which is viable precisely because the deterministic witness is small (the largest blob is 1.08 MB) and the replay is cheap.

Provenance and data standards. PROV-O W3C Provenance Working Group (2013) and Darwin Core Wieczorek et al (2012) describe lineage and occurrence records; FAIR Wilkinson et al (2016) governs discoverability; CARE Carroll et al (2020) governs consent for data about communities. Our consent gate operationalizes a CARE-style principle — collective benefit, authority to control — as an executable admission rule rather than a documentation requirement.

Ecological sensing and MRV. Open acoustic hardware Hill et al (2018); Pearce (2012), bird-sound classifiers Kahl et al (2021), and ecoacoustic indices Sueur and Farina (2015) supply the sensing frontier; MRV criteria (e.g. SAGED Ford et al (2024)) define what counts as credible evidence. The living-systems lane claims motivating our gate design — airborne plant stress sounds Khait et al (2023), biofilm ion-channel waves Prindle et al (2015), biofilm-to-biofilm coupling Liu et al (2017), electrical cell attraction Humphries et al (2017), and contactless conductivity sensing Kubáň and Hauser (2009) — are catalogued per-claim in Appendix C. This paper claims none of them as inputs; they motivate why the consent lane exists.

Positioning. Table 1 summarizes the verification landscape.

3 The protocol

3.1 Lanes and gates

Every operation in the commons is assigned a lane, and every lane satisfies exactly one gate (formal specification in Appendix A): *sense* (a sensor reports an observation), *fleet* (siblings corroborate an observation), and the refused *compute* lane (claiming an organism computed) are *evidence-gated*: admission requires a verifiable evidence claim. *emit* (drive a transducer toward a living system) and *store* (write on a guardian-controlled node) are *consent-gated*: admission requires a consent tuple ⟨guardian, operator, action-class, policy-hash⟩ that verifies at admission. *hop* (relaying through living tissue) and any claim of in-organism computation or storage are refused unconditionally, as is promotion of a federated receipt to local quorum status. A refusal is a signed, content-addressed protocol value — it is *data*, not silence.

3.2 Witnesses, quorum, receipts

A sensing input is rasterized into a feature vector ⟨confidence, event-rate, env-anomaly, fleet-agreement⟩ (Section 4) and replayed through each admitted connectome witness. Each witness emits a verdict: (action $\in \{-1, 0, +1\}$, confidence, spike-count, digest). A quorum is ≥ 3 distinct connectomes agreeing on action; quorum output folds into a receipt addressed by BLAKE2s SAID — self-authenticating: the address is the content's hash, so any holder can verify integrity without trusting the store. Receipts are the unit mutual credit settles against: a sensed detection contributes to credit only through a witnessed, receipted record, never an unwitnessed claim.

3.3 Custody

Guardians hold custody of deployed nodes; operators hold keys for service accounts; subjects (living systems) earn credit. Consent is consent of the *custodian*, never attributed to the organism — the model is explicit about this boundary and the paper makes no claim of organism agency.

4 Connectome witnesses: methods

4.1 Corpus

WiredReplay-1 contains fifteen wiring diagrams spanning two resolution classes, deliberately not blurred: cellular EM reconstructions (302–40,000 neurons) and mesoscale parcellation graphs (49–242 neurons). Table 2 lists the corpus; Appendix A gives the normative packed format. Blob sizes are 4.9 KB–1.08 MB (packed int4-CSR): the entire cellular medulla wiring fits in a megabyte.

4.2 Packed representation

Graphs are packed as int4 CSR (indptr deltas, u16 indices, 4-bit two's-complement weights, two per byte) in $\leq$24,575-byte chunks with a JSON sidecar — a derivative of

Table 2 WiredReplay-1 corpus. Resolution is stated per dataset; "mesoscale" graphs are interareal parcellations, not cellular EM. Janelia-derived blobs (hemibrain, medulla, larva) are CC-BY redistributable; cellular-scale rows are "published-plausible" (matches to published connectomes by size and count); the five mesoscale rows drosophila, human, mouse, rat and macaque carry *provenance-pending* status — their upstream sources could not be re-identified from the packed artifacts, and the corpus flags them accordingly rather than asserting an unverified source. Determinism results do not depend on provenance; dataset licensing status is tracked in `paper/datasets.md`.

connectome	neurons	synapses	resolution
medulla	40,000	399,950	cellular EM
hemibrain	21,636	216,310	cellular EM
platynereis	5,000	49,991	cellular EM
larva	3,016	30,149	cellular EM
mouse_retina	1,124	90,009	cellular EM
celegans_male	575	5,305	cellular EM
celegans_herm	453	4,878	cellular EM
celegans	302	3,009	cellular EM
human	234	7,076	mesoscale
macaque_modha	242	4,090	mesoscale
ciona	205	2,902	cellular EM
mouse	112	6,542	mesoscale
macaque	93	1,692	mesoscale
rat	73	1,923	mesoscale
drosophila	49	1,950	mesoscale

the MIT-licensed FlyBrain pack format. Packing is the efficiency instrument: weights are quantized to nibbles, addresses compressed to per-neuron deltas.

4.3 Determinism contract

A conformant simulator must satisfy, end-to-end: integer arithmetic throughout (Q24 fixed-point, floor division, right-shift rescale — no floating point anywhere Goldberg (1991)); no RNG — per-neuron input variation derives deterministically via BLAKE2s from the neuron index (encoder `lif-sensing-v1`, Appendix A); a fixed step count per blob; and digests defined over the exact byte serialization. Determinism is what makes *replay* the receipt: equal digests from independent implementations is an astronomically-unlikely coincidence, not agreement by tolerance.

4.4 Replay protocol

Given a verdict ⟨connectome, input-SAID, features, steps, action, confidence, spike-count, digest⟩, a verifier re-rasterizes the features, re-simulates, and compares digests byte-for-byte. The deployed worker exposes this as a public `/verify` endpoint; the corpus vectors replay the same way offline.

5 Results

5.1 E5: relay with topology-unique signatures

The pre-registered E5 criterion (an input relays through all connectomes, each producing a distinct signature) **passed**: 15/15 relays, 15 distinct BLAKE2s signatures, replayed identically by the Python reference runner and the TypeScript worker (cross-implementation). The signatures are fingerprint-grade — two different wirings of the same input produce different digests — which is what lets a witnessed receipt distinguish *which* wiring diagram witnessed a claim.

5.2 Cross-implementation parity

The independent implementations agree on 930/930 parity checks (feature grids over all 15 connectomes): identical actions, confidences, and digests. Divergence observed in development — a documented keccak→blake2s variation-source change between the runner and its on-chain ancestor — is a named encoder version (`lif-sensing-v1`), demonstrating the corpus's intended use: differences are caught at digest granularity and versioned.

5.3 Protocol and deployment

The five e2e suites (connectome witness, federation, refused lanes, sensing cell, write lanes) total 121 checks, all passing under GNU Guile 3.0. The live worker passed 71/71 checks against the production deployment at templeearth.cc, including a live quorum→verdict round-trip and public `/verify` replay of a production receipt ("live now" tier; logs in `paper/artifacts/`).

5.4 Efficiency, honestly

Figure 1 and Table 3 report the two benchmark legs on the same host (fresh runs in `paper/artifacts/`; earlier measured-tier runs appear in `docs/EFFICIENCY.md`). Fresh-run joules use a CPU-time×15 W proxy and are labelled *estimated*.

Large wiring (medulla, 40k neurons). The connectome leg: 882,532 ops over 1,079,875 packed bytes, 2.05 ms, 3.07×10^{-2} J (est.). Against same-host baselines: dense fp32 1.6×10^{9} ops / 6.4 GB / 115.8 ms / 38.1 J (est.); sparse CSR 399,950 ops / 2.4 MB / 1.34 ms / 0.27 J (est.). The connectome is ∼56× faster and ∼1,200× cheaper than dense; it sits between dense and sparse exactly where its op count predicts.

The control is the finding. A degree-matched shuffled medulla (same neurons, same degree sequence, randomized targets) produces statistically identical timing and energy (2.06 ms, 3.08×10^{-2} J) — *the saving is the sparse packed representation, not the biological topology.* We report this deliberately: what the biological wiring contributes is the *signature* (a unique, replayable fingerprint of a real evolved wiring), not the speed.

Small wiring (celegans, 302 neurons). The honest loss: our leg is *slower* than both baselines (0.120 ms vs 0.007 dense / 0.008 sparse) — Python+NumPy overhead dominates below a scale threshold. There is a crossover (roughly, the regime where

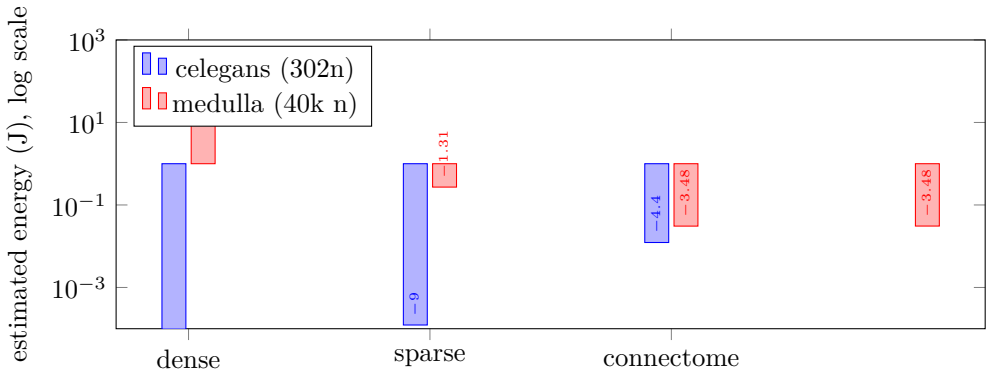


Fig. 1 Estimated energy per simulation run (CPU-time×15 W proxy). At cellular scale the connectome leg is ∼3 orders of magnitude cheaper than dense; at 302 neurons it loses to both baselines — reported as the scale-limit honesty case. The degree-matched shuffled control overlaps the real wiring, showing the saving is representation, not topology.

Table 3 Benchmark legs, fresh runs (tier: *estimated*). Ops counts are semantic synaptic updates vs. MACs; "shuffled" is the degree-matched control. Estimated joules are CPU-time×15 W — the multithreaded BLAS baselines burn substantially more CPU-time than wall-time (dense: 2,540 ms CPU / 115.8 ms wall; sparse: 18.0 ms CPU / 1.34 ms wall), while the connectome leg is single-threaded (2.04 ms CPU / 2.05 ms wall). Wall-time figures therefore tell a different story than the energy column — sparse CSR is marginally faster on the wall clock but ∼9× more expensive in CPU-joules.

wiring	leg	ops	bytes	est. J
medulla	dense fp32	1.6×10^9	6.4×10^9	38.1
	sparse CSR	399,950	2,399,700	0.271
	connectome	882,532	1,079,875	0.0307
	shuffled	882,532	1,079,875	0.0308
celegans	dense fp32	91,204	364,816	9.8×10^{-5}
	sparse CSR	3,009	18,054	1.2×10^{-4}
	connectome	35,346	7,825	1.2×10^{-2}

$O(\text{steps}\times\text{synapses})$ sparse work beats the $O(n^2)$ dense multiply and where interpreter overhead is amortized); medulla is past it, celegans is not. Both legs, including the loss, ship in the corpus vectors.

6 Discussion

6.1 What the witnesses are — and are not

A witness signature certifies that *this wiring diagram, under this input, produced this raster*. Quorumed, it supports a claim that a sensed observation was processed through

a documented, replayable pipeline. It does not certify that an organism computed, relayed, or consented — the connectome is a preserved wiring *used as* computation; the organism is not claimed as the computer. This distinction is enforced in the protocol's lane table (living-system compute is refused) and in the claim labels.

6.2 The living cloud, honestly bounded

The architecture is designed so that a future E0+ deployment — sensors reading living systems, consent-gated emission, receipts crediting subjects — plugs into the receipt machinery demonstrated here without protocol changes. What this paper demonstrates is the verifiable spine: provenance that degrades into refusal rather than into trust, and a witness layer cheap enough to replay anywhere. Field validation is explicitly out of scope (Section 6.4).

6.3 Threat model

Appendix B details adversary classes; the load-bearing property is that *forgery requires either breaking BLAKE2s or defeating determinism*, while the honest-participant failure modes (mis-rasterized input, wrong blob version) are detectable at digest granularity. The residual trust point is the sensor itself — identical in kind to every remote-sensing system, and why the evidence gates exist upstream of the witnesses.

6.4 Limitations

(i) No field deployment: E0–E6 are pre-registered; none of the "living" lanes has produced a witnessed real-world receipt. (ii) Estimated joules: fresh benchmark runs use a power proxy; the measured-tier numbers in `docs/EFFICIENCY.md` come from RAPL-unlocked runs and are labelled accordingly. (iii) The witness layer validates *computation*, not *biology*: connectome verdicts carry no claim about cognition, ecology, or the health of the source organism. (iv) Sole-author project; the corpus is young. (v) Mesoscale and cellular graphs are mixed in one corpus deliberately — resolution is declared per row, and any use must respect it.

7 Availability

Code (AGPL-3.0): https://github.com/TerexitariusStomp/living-resources — `tools/connectome/`, `workers/compute-api/`, `tests/e2e/`. Corpus + vectors + spec: `paper/corpus/`, archived at https://doi.org/10.5281/zenodo.23681503 (release `paper-v1`). Reproduction commands: `paper/REPRODUCE.md`. Deployed API: https://templeearth.cc (`/verify` is a live replay endpoint). Claim ledger: `docs/SCIENCE.md`; pre-registered experiments: `docs/EXPERIMENTS.md`.

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Ethics and consent

No living system was instrumented, stimulated, or emitted at for this work; all connectome data are prior published reconstructions. Guardian-consent machinery described here is implemented software, unexercised in the field. No animal experiments were performed.

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Appendix A Formal lane and receipt specification

Lanes. $\mathcal{L} = \{\mathsf{sense}, \mathsf{fleet}, \mathsf{emit}, \mathsf{store}, \mathsf{hop}, \mathsf{compute}\}$. Gate map g: $\{\mathsf{sense}, \mathsf{fleet}\} \mapsto$evidence; $\{\mathsf{emit}, \mathsf{store}\} \mapsto$consent; $\{\mathsf{hop}, \mathsf{compute}\} \mapsto$refuse. Admission: $\mathsf{admit}(op) = \mathsf{admitted}(x)$ iff op.lane's gate verifies, else $\mathsf{refused}(r)$ — a signed value. **Consent tuple.** $C = \langle g, o, a, p \rangle$: guardian, operator, action class, policy hash; verified at admission, expiring, non-transferable. **Evidence claim.** $E = \langle s, \tau, \sigma \rangle$: source id, claim type, attestation; verifiable means checkable by a third party. **Witness verdict.** $W = \langle c, said_{in}, f, k, a, \gamma, s, d \rangle$: connectome, input SAID, feature vector, steps, action $\in \{-1, 0, +1\}$, confidence, spike count, digest. **Quorum.** ≥ 3 distinct c agreeing on a; output folds to $receipt = \mathsf{said}(\{W_i\}, policy, context)$ where $\mathsf{said}(x) =$"blake2s:" $||$ blake2s(canon(x)). **Encoder** `lif-sensing-v1` and blob layout are the normative text of `paper/corpus/SPEC.md` (reproduced there in full); constants: Q24 fixed-point, decay = 13,743,895, tonic 2,359,296, threshold 16,777,216, gain 25,165,824, weight-scale 7,190,235, variation-mod 1,677,720; rasterizer splits the population into affirm/refute halves driven by confidence, anomaly, agreement, and rate (equations in SPEC.md §4).

Appendix B Threat model

Adversary classes. (a) *False detection*: a node claims an event that did not occur. Mitigation: the receipt binds the input SAID to a feature vector whose sources are themselves receipted; a forged claim needs forged upstream receipts — the same problem all MRV systems face, surfaced rather than hidden. (b) *Claimed replay*: asserting a digest without simulating. Defeated by public re-execution — the verification is the re-run. (c) *Implementation divergence*: a verifier drifts from the reference semantics. WiredReplay-1 vectors catch this at digest granularity. (d) *Quorum stuffing*: multiple

verdicts from one wiring. Quorum counts *distinct connectome ids*. (e) *Consent bypass*: emitting without a tuple, or a tuple minted by the operator alone. The consent gate requires the guardian principal; emission is additionally duty-cycled by policy hash. (f) *Federated laundering*: importing an external receipt as local quorum. Federation marks provenance; refused by the lane table. **Residual trust**: the physical sensor and the guardian's key custody — universal to remote sensing, and the reason evidence gates sit upstream of witnesses.

Appendix C Claim table

Condensed from `docs/SCIENCE.md` (authoritative):

- *Proven by science*: plant stress ultrasounds airborne + informative Khait et al (2023); biofilm K^+ waves Prindle et al (2015); biofilm coupling Liu et al (2017); electrical cell attraction Humphries et al (2017); reef acoustics track biodiversity Sueur and Farina (2015); C4D contactless conductivity Kubáň and Hauser (2009).
- *Built / live*: LIF-over-connectome replay with digest signatures (live: 15/15, parity 930/930); consent-gated emission stack (built, unexercised); quorum→receipt (live, 71/71 e2e); federation with provenance (built); mutual-credit settlement (built, zero credited detections to date).
- *Measured*: medulla crossover vs. dense fp32 (EFFICIENCY.md, RAPL-unlocked runs); fresh-run figures here are *estimated* tier. *Estimated*: all fresh-run joules. *Refused*: organism-as-computer, hop through living tissue, in-organism storage, ultrasound-as-instruction-set, unconsented emission, federated-to-quorum promotion.

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